**Project Title: Meesho Jewelry Marketplace Database**

**SALES AND CUSTOMER ANALYTICS SYSTEM**

## 1. Objectives

1. Apply aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Identify top customers based on their purchase amount.
4. Identify the best-selling products.
5. Perform category-wise sales analysis.

## 2 .Applying Aggregate Functions

### 2.1 COUNT() – Counts the total number of orders.

```
SQL> SELECT COUNT(*) FROM ORDERS;

  COUNT(*)
----------
         5
```

### 2.2 SUM() – Calculates the total sales amount.

```
SQL> SELECT SUM(TOTAL_AMOUNT) FROM ORDERS;

SUM(TOTAL_AMOUNT)
-----------------
             5994
```

### 2.3 AVG() – Calculates the average order amount.

```
SQL> SELECT AVG(TOTAL_AMOUNT) FROM ORDERS;

AVG(TOTAL_AMOUNT)
-----------------
           1198.8
```

### 2.4 MIN() – Finds the minimum product price.

```
SQL> SELECT MIN(PRICE) FROM PRODUCT;

MIN(PRICE)
----------
       599
```

### 2.5 MAX() – Finds the maximum product price.

```
SQL> SELECT MAX(PRICE) FROM PRODUCT;

MAX(PRICE)
----------
      1599
```

### 3. Generate Total Sales Reports

Display the total sales amount and total number of orders.

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  4      AVG(TOTAL_AMOUNT) AS AVERAGE_SALES
  5  FROM ORDERS;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES
------------ ----------- -------------
           5        5994        1198.8
```

### 4. Top Customers Based on Purchase Amount

Display customers according to their total purchase amount, from highest to lowest.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  INNER JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  9  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
          1 Priya                                                        2797
          2 Anu                                                          1599
          3 Divya                                                         999
          5 Meena                                                         599
```

## 5. Identify Best-Selling Products

Find products based on the total quantity sold.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  INNER JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID PRODUCT_NAME                                                                                          TOTAL_QUANTITY_SOLD
---------- ------------------------------------------------------------------------------------------------------ -------------------
       201 Gold Plated Necklace                                                                                                     2
       202 Diamond Earrings                                                                                                         1
       205 Stone Ring                                                                                                               1
       203 Pearl Bracelet                                                                                                           1
```

## 6. Category-Wise Sales Analysis

This query analyses product sales category-wise and calculates the total quantity sold in each category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM CATEGORY C
  6  INNER JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  INNER JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

CATEGORY_ID CATEGORY_NAME                                                                                          TOTAL_QUANTITY_SOLD
----------- ---------------------------------------------------------------------------------------------------- -------------------
          2 Necklaces                                                                                                            2
          1 Rings                                                                                                                1
          3 Earrings                                                                                                             1
          5 Bangles                                                                                                              1
```